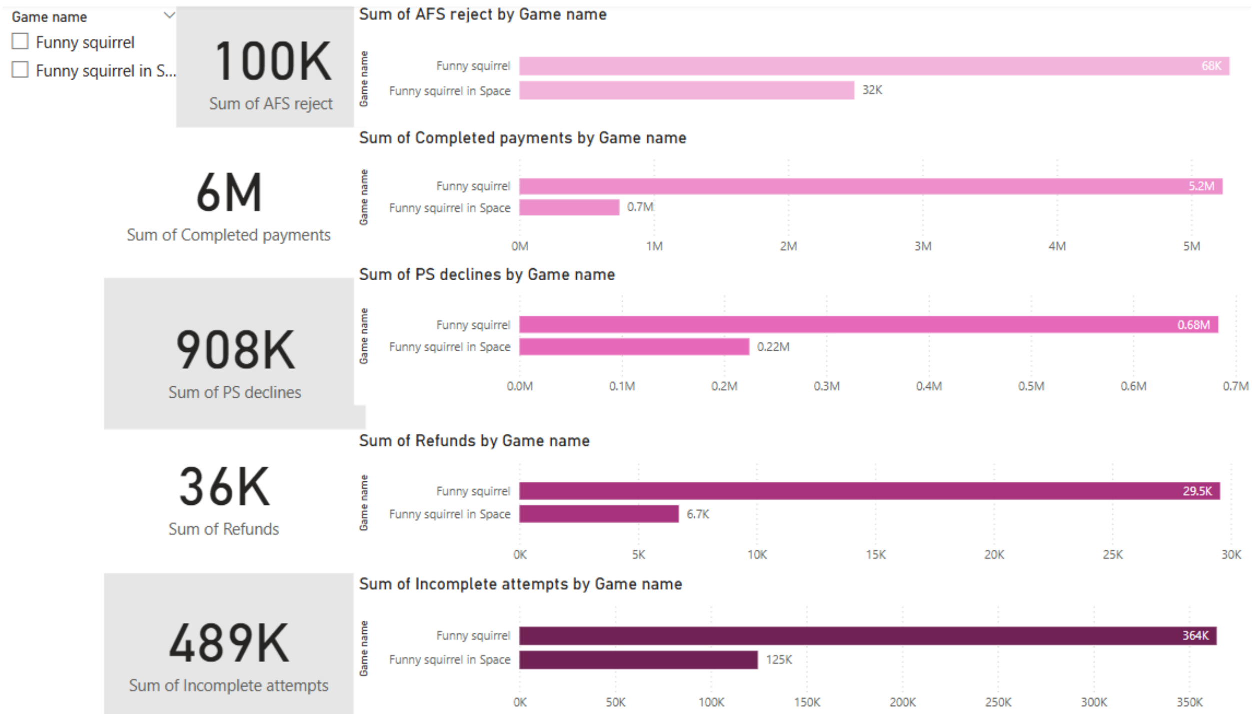
Task 4

There are several sections in this Task.

1. Game-Level Payment Flow Overview



This dashboard offers a comprehensive breakdown of payment performance across both games: *Funny Squirrel* and *Funny Squirrel in Space*. It visualizes the full transaction journey from incomplete attempts to rejections, successful payments, and final refunds.

☑ Key Performance Metrics:

Funny Squirrel in Space	Funny Squirrel	Metric
743K	5.2M	Completed Payments
224K	683K	PS Declines
32K	68K	AFS Rejects
6.7K	29.5K	Refunds
125K	364K	Incomplete Attempts

☑ Insights and Observations:

1. Transaction Volume Dominance:

- Funny Squirrel clearly leads in total volume, processing over 5 million successful transactions, almost 7× more than its sequel.

2. System Rejections:

- Despite its volume, Funny Squirrel experiences significantly more PS declines and AFS rejects.
- However, rejection rates relative to completions are lower in Funny Squirrel in Space, indicating greater efficiency in handling transactions.

3. Refunds:

- Although Funny Squirrel sees more total refunds (29.5K), the refund rate is only slightly lower than that of Funny Squirrel in Space due to higher volume.
- The relatively low absolute number of refunds in both games is a positive signal.

4. Incomplete Attempts:

- Funny Squirrel also sees almost 3× more incomplete attempts, which may be tied to its broader reach, user demographics, or friction points in its payment UX.

☑ Strategic Interpretation:

- Funny Squirrel in Space demonstrates a cleaner transaction funnel, with fewer rejections and refunds per transaction, despite lower total volume.

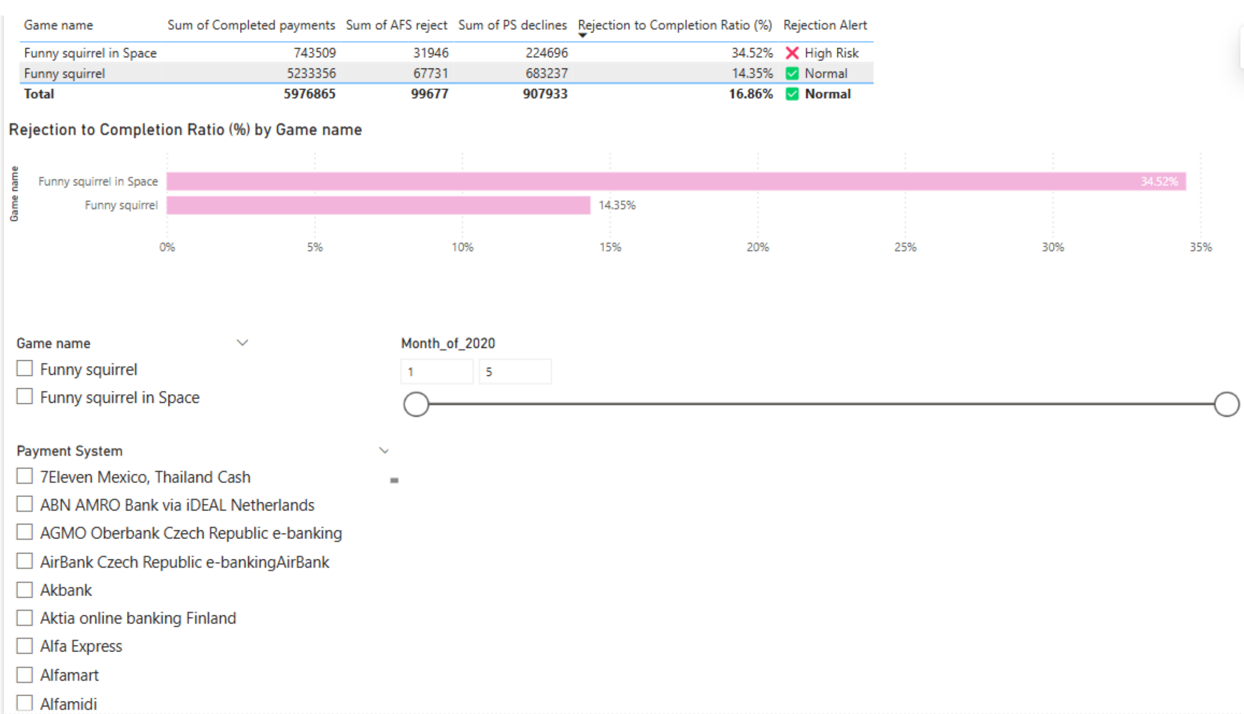
- Funny Squirrel handles high traffic but may require further optimization in error handling, user guidance, and funnel drop-off points.

☑ Recommendations:

- Conduct a conversion funnel audit for Funny Squirrel to identify and reduce incomplete or rejected transactions.
- Use Funny Squirrel in Space as a benchmark for streamlined payment flow.
- Prioritize refinement of the payment UI/UX and retry logic in the high-traffic game to improve retention and revenue.

☑ This holistic comparison provides actionable insights into payment process efficiency, technical reliability, and user experience quality across products.

2. Game-Level Rejection Ratio Analysis



This slide focuses on evaluating the reliability of each game's payment experience by analyzing the Rejection to Completion Ratio (%). This metric reflects the share of failed payment attempts (AFS + PS rejections) in relation to successful transactions.

☑ Key Observations:

Funny Squirrel in Space demonstrates a high rejection ratio of 34.52%, which has triggered a "High Risk" alert.

This means that over one-third of all attempted transactions failed for this game.

It raises significant concerns around the payment experience and system compatibility for this title.

Root causes could include technical misalignment, payment gateway issues, or user behavior anomalies.

Funny Squirrel, in contrast, has a rejection rate of 14.35%, which remains within a normal range.

This suggests a more stable and reliable payment process for this game, despite its significantly larger transaction volume.

The global average rejection ratio stands at 16.86%, which positions Funny Squirrel in Space considerably above the benchmark, requiring immediate review.

☑ Strategic Interpretation:

A rejection rate above 30% is generally considered critical in digital payments, especially for products with global audiences.

High rejection rates not only reduce revenue conversion but also frustrate users, potentially leading to churn and negative feedback.

☑ Recommendations:

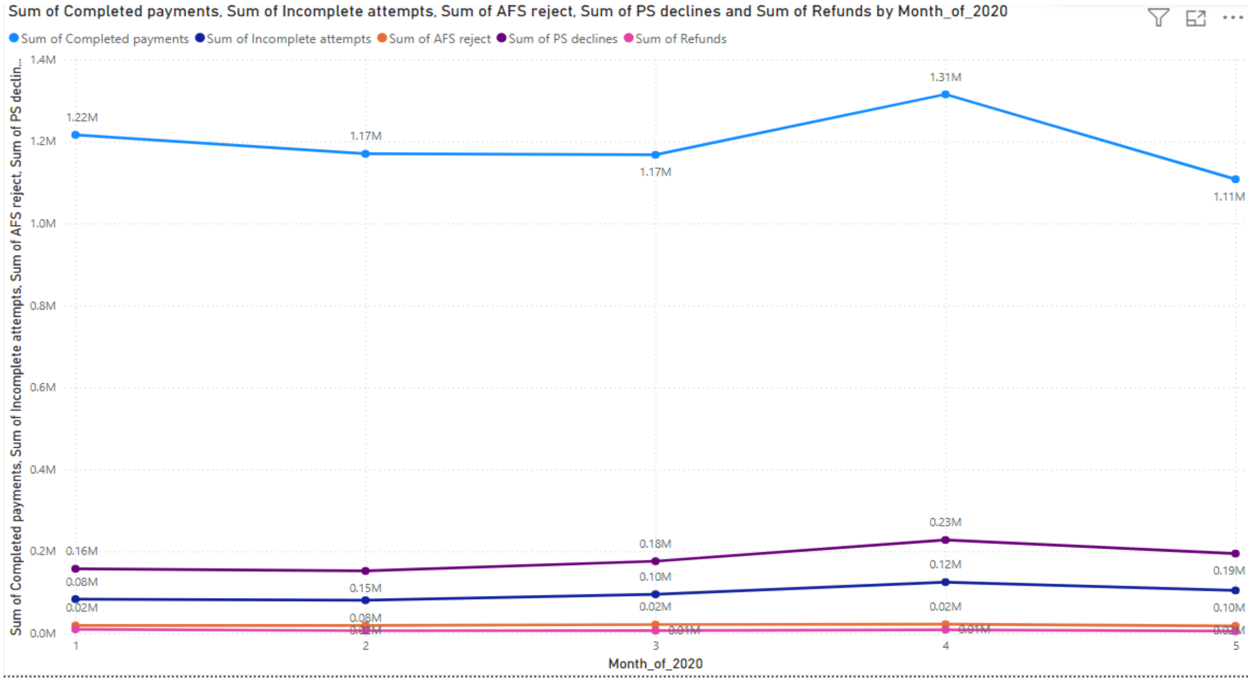
Conduct a technical audit of Funny Squirrel in Space's payment flow, particularly for top-rejecting systems or countries.

Evaluate integration and error-handling mechanisms across payment providers associated with this game.

Consider implementing fallback payment options or improved user guidance in case of initial failure.

☑This dashboard effectively pinpoints areas of operational risk at the game level, enabling the product and engineering teams to focus resources where user experience and revenue are most at stake.

3. Monthly Transaction Flow Trend – From Attempt to Refund



This time-series visual presents the monthly dynamics of user payment behavior and system response throughout the first five months of 2020. The chart tracks five core metrics:

- Completed Payments
- Incomplete Attempts
- AFS Rejects
- PS Declines
- Refunds

☑ Key Observations:

1. Completed Payments – Stable with Peak in Month 4

- Total completions remain stable from Month 1–3 (Approximately 1.17M average)
- A clear peak at 1.31M in Month 4, likely due to a promotional or seasonal campaign
- Month 5 shows a dip to 1.11M, returning to baseline

2. PS Declines (Purple Line)

- A gradual increase from 160K in Month 1 to a peak of 230K in Month 4, tracking closely with traffic volume

- This mirrors the completed payments pattern — high usage months also show more failures

3. Incomplete Attempts (Blue-Purple)

- Slight but steady rise, peaking at 120K in Month 4, suggesting users may have been confused or abandoned the payment process more often during that time

4. AFS Rejects (Orange Line)

- Flat and low throughout (20K), no notable variation — suggesting consistent fraud detection behavior

5. Refunds (Red Line)

- Remain very low overall (~6K per month)
- Slight increase in Month 4 but not significant — implying user satisfaction didn't drop sharply even during volume spikes

☑ Interpretation & Implications:

- Month 4 is a traffic and performance hotspot — both success and failure metrics peaked.
- System appears stable, but PS declines and incomplete attempts should be monitored during traffic surges.
- Refund volume staying low suggests users generally trust the system and issues may be technical, not user-driven.

☑ Recommendations:

- Conduct deeper technical monitoring and load testing in high-traffic periods like Month 4
- Improve payment UX to reduce incomplete attempts during busy cycles
- Maintain consistent AFS settings but explore adaptive fraud models for dynamic volume

☑ This trend view allows stakeholders to understand when and why the system struggles under load—and respond proactively to protect both revenue and user experience.

4. Country-Level Summary – Key Insights

1. Top Completed Payments:

- United States, Australia, and South Korea lead in completed payments with 640K+, 550K+, and 480K+ respectively.
- These are the primary revenue-generating regions.
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2. Highest PS Declines:

- Turkey, Germany, Brazil, and South Korea show the most PS declines (up to 78K), indicating payment processing issues in these markets.

3. AFS Rejects (Anti-Fraud Triggers):

- United States tops with 20.9K AFS rejections, followed by Turkey and Saudi Arabia.
- This may reflect aggressive fraud policies or high-risk transaction patterns.

☑ Recommendation: Optimize payment systems in TR, DE, and BR to reduce friction. Review fraud detection thresholds in US to avoid unnecessary blocks.

5. Country Success Rates Table

☑ Highest Success Rates:

BE (90.86%), NZ (97.92%), NO (97.91%), and CA (97.48%) show excellent reliability in payment flow.

☑ Lowest Performing Countries:

CL (71.5%), AR (86%), BR (80.2%), IT (83.1%) — all have high failure rates (≥ 25–30%), needing further investigation.

US US leads in both total transactions (741K) and average success (638K) with 96.4% success rate → most stable and profitable market.

☑ Overall Metrics:

- Total transactions: 7.47M
- Average global success rate: 92.44%
- Global failure rate: 13%

Additionally, several low-volume countries such as TM, KS, AN, PM, RW, VU, NU, BJ, MW, NE, and SL achieved a perfect 100% success rate, highlighting technical reliability—though their contribution to total volume remains minimal.

6. Rejection Volume Analysis – Transactions vs Success

Game name	Sum of Completed payments	Sum of AFS reject	Sum of PS declines	Rejections Greater Than Completed
Funny squirrel	5233356	67731	683237	No
Funny squirrel in Space	743509	31946	224696	No
Total	**5976865**	**99677**	**907933**	**No**

User country	Sum of Completed payments	Sum of AFS reject	Sum of PS declines	Rejections Greater Than Completed	Country Excess Rejection Flag
AS	0	1	1	Yes	YES
BF	3	1	3	Yes	YES
BI	0	4	1	Yes	YES
BJ	1	3	3	Yes	YES
BQ	8	2	7	Yes	YES
CG	0	2	1	Yes	YES
Total	**5976865**	**99677**	**907933**	**No**	**NO**

39

Countries with Excess Rejections

Payment System	Sum of Completed payments	Sum of AFS reject	Sum of PS declines	Rejections Greater Than Completed
7Eleven Mexico, Thailand Cash	59	1	879	Yes
AGMO Oberbank Czech Republic e-banking	0	0	1	Yes
AirBank Czech Republic e-bankingAirBank	80	0	86	Yes
Akbank	31	14	452	Yes
Alfa Express	0	1	5	Yes
Alfamart	33	10	437	Yes
Total	**5976865**	**99677**	**907933**	**No**

161

Systems with Excess Rejections

Month_of_2020	Sum of Completed payments	Sum of AFS reject	Sum of PS declines	Rejections Greater Than Completed
1	1216316	19118	157492	No
2	1169985	19025	152222	No
3	1167575	21375	175936	No
4	1314985	22109	227758	No
5	1108004	18050	194525	No
Total	**5976865**	**99677**	**907933**	**No**

This section evaluates the proportion of failed transactions (AFS rejections and PS declines) relative to successful, completed payments. The focus is to identify countries, payment systems, and time periods where failures outnumber completions—an indicator of systemic risk, technical malfunction, or poor user experience.

☑ Overall Summary:

Out of all records, 39 countries and 161 payment systems exhibit a higher number of rejections than completed transactions.

This indicates either:

- Technical instability in the payment process
- Fraud prevention overreach (AFS)
- Mismatch between the payment system and user behavior
- Extremely low transaction volume magnifying rejection ratios

☑ Country-Level Insights:

Countries such as AS, BF, BI, BJ, BQ, and CG show rejection rates exceeding completion rates.

In most of these cases, the number of completed payments is extremely low or zero, suggesting a possible market mismatch or accessibility issue.

These flags help prioritize which regions may require payment flow adjustment or infrastructure review.

☑ Payment System Risks:

Several payment systems—including 7Eleven Mexico, Akbank, Alfa Express, and AirBank Czech Republic—have more failures than successful transactions.

In systems with very low volume (e.g., <100 transactions), even a few failures lead to extreme ratios. These should still be monitored as part of a long-tail risk strategy.

☑ Time-Based Observation:

Across all months analyzed (Months 1–5), rejection volume remained below completed transactions—indicating that the overall system remains stable across time, even though localized failures exist.

☑ Strategic Implications:

High rejection volume compared to completions may result in:

- Loss of user trust
- Increased support costs
- Lower conversion rates

These patterns should trigger a deeper investigation into the payment routing logic, fraud rules, and UX friction points.

☑ Recommended Actions:

Audit the top flagged payment systems and regions with consistent rejection dominance.

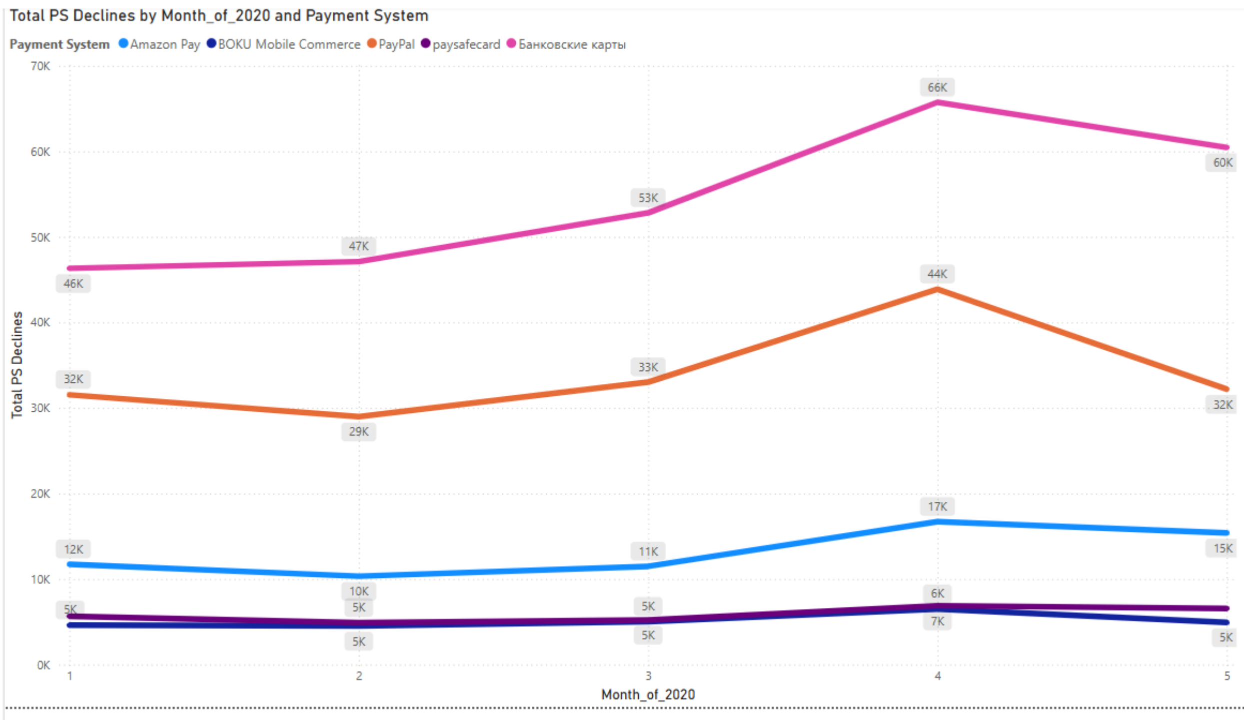
Disable or adjust systems with no successful transaction history.

Reassess anti-fraud logic thresholds in low-volume geographies.

Consider fallback payment options or rerouting strategies in high-failure areas.

☑ This diagnostic analysis brings operational clarity and identifies payment process inefficiencies at scale, helping stakeholders to focus efforts where failure is frequent but volume is hidden.

7. Monthly Trend of Payment System Declines



This visual represents the trend of payment system (PS) declines over the first five months of 2020, segmented by the most used payment providers.

☑ Key Observations:

- Банковские карты (Bank Cards) consistently reported the highest PS decline volume, peaking at 66K in Month 4.
- This steady increase suggests a possible system overload or recurring incompatibility issues during the mid-period.
- Even though it dipped slightly in Month 5, the decline volume remains significantly higher than all other systems.

- Paysafecard follows as the second-highest with 44K declines in Month 4.
- Interestingly, its trend closely mirrors Bank Cards, indicating a broader ecosystem-related issue or seasonal pattern in user behavior.
- PayPal shows a consistent but significantly lower trend, ranging from 10K to 17K, peaking in Month 4.

This indicates a reliable system with manageable rejection rates.

Amazon Pay, BOKU Mobile Commerce, and AirBank Czech Republic exhibit very low decline volumes (mostly around 5–7K), with minimal variation over time.

These systems can be considered relatively stable.

☑ Month 4 (April) as a Critical Spike:

Across almost all systems, Month 4 shows a noticeable spike in PS declines.

Possible contributing factors:

- Peak traffic due to seasonal in-game events or marketing campaigns
- Technical bottlenecks or fraud rule misfires
- Behavioral anomalies such as bulk attempts or bots

☑ Strategic Recommendations:

Deep-dive into Month 4 logs and performance metrics for Bank Cards and Paysafecard to uncover root causes.

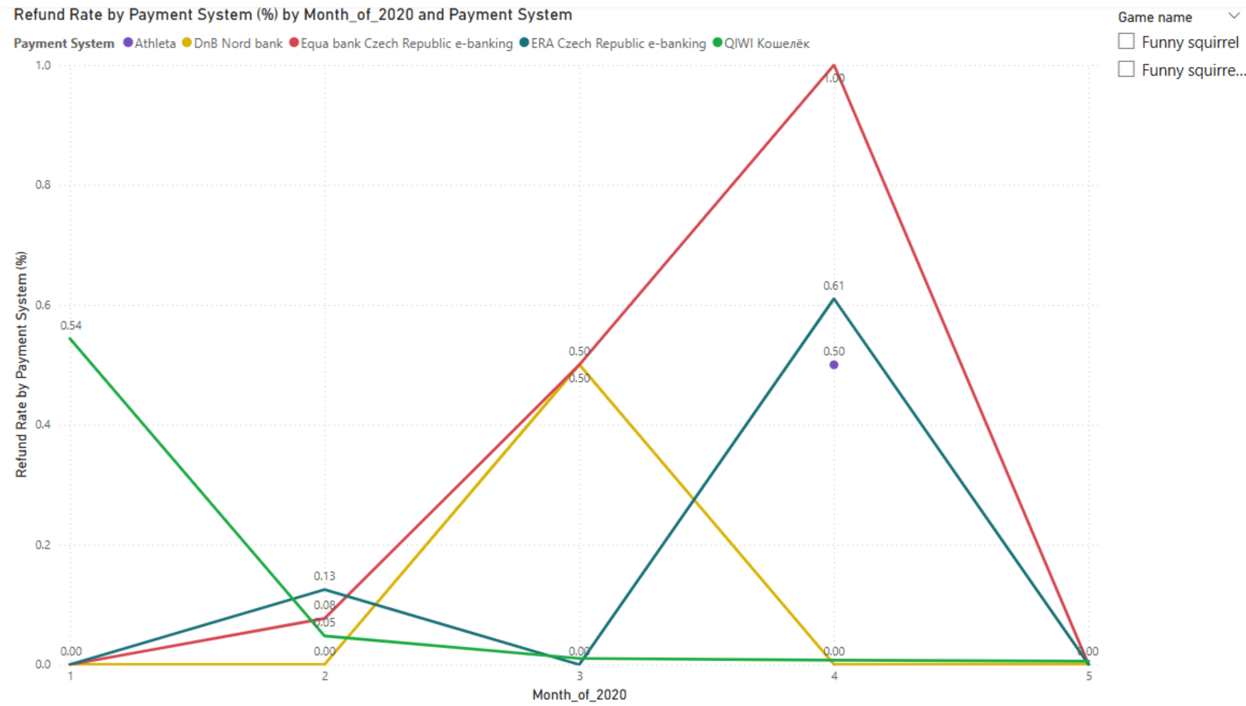
Introduce proactive load testing and error tracking ahead of peak activity months.

Evaluate alternative routing or fallback methods for high-risk systems during seasonal traffic peaks.

Maintain and promote stable systems like PayPal, Amazon Pay, and BOKU as preferred default options in high-risk geographies.

☑ This trend-based PS decline analysis offers valuable insight into system reliability, helping payment and product teams optimize routing logic and customer experience during high-volume cycles.

8. Refund Rate Trend by Payment System – Monthly Breakdown



This chart illustrates the Refund Rate (%) per payment system across the first five months of 2020. It captures critical spikes that reflect transactional dissatisfaction or systemic issues.

☑ Key Observations:

- Equa Bank Czech Republic e-banking reported the highest refund rate of 1.00% in Month 4, a sharp spike from 0.5% in Month 3.
- This pattern signals a consistent issue escalating over time, likely related to system error, policy change, or user experience friction.
- ERA Czech Republic e-banking also peaked in Month 4 with a refund rate of 0.61%, indicating that multiple Czech-based systems faced user pushback in the same period.
- DnB Nord Bank showed a refund rate of 0.50% in both Month 3 and Month 4, reinforcing the concern around refund behavior in Central European payment infrastructure during this window.
- QIWI (Кошелёк) showed a refund rate of 0.54% in Month 1, followed by a drop to zero in subsequent months.
- This anomaly may be due to a short-term disruption or correction in the early phase of the year.

- Athleta, despite appearing in the graph, remained stable with minimal or no refunds during this period.

☑ Month 4 Identified as a Critical Period

Multiple systems peaked in refund ratio in Month 4 (April), indicating a broad systemic friction point.

This could relate to:

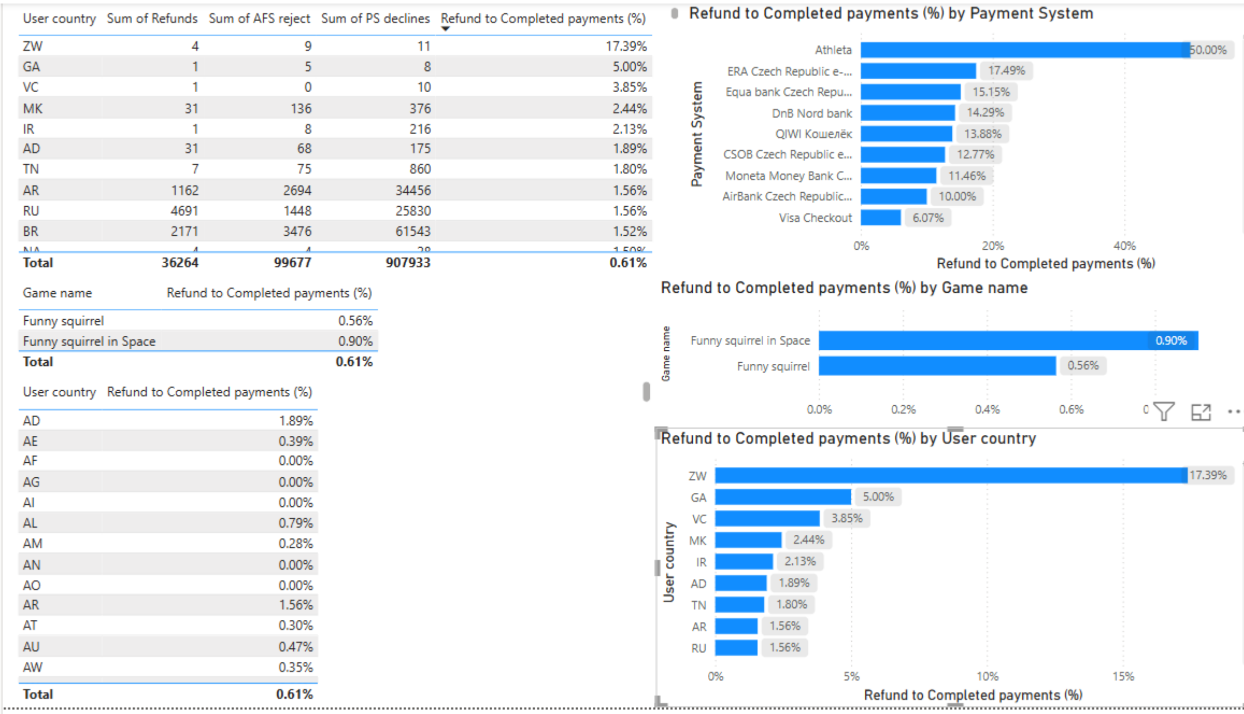
- Seasonal spikes in user traffic (e.g., promotional events)
- Updates or outages in banking systems
- Shifts in fraud detection or refund policy

☑ Strategic Recommendations:

- Investigate backend logs and error reports for Equa and ERA Czech systems in April.
- Review refund handling mechanisms and thresholds for all providers peaking in Month 3–4.
- Consider communication adjustments or extra support for users during system anomalies.
- Promote more stable systems during high-risk months to avoid large-scale refund surges.

☑ This analysis provides time-sensitive insight into user trust, payment system integrity, and product experience stability—key drivers of retention and revenue health.

9. Refund Behavior Analysis – Executive Insight



User country	Sum of Refunds	Sum of AFS reject	Sum of PS declines	Refund to Completed payments (%)
ZW	4	9	11	17.39%
GA	1	5	8	5.00%
VC	1	0	10	3.85%
MK	31	136	376	2.44%
IR	1	8	216	2.13%
AD	31	68	175	1.89%
TN	7	75	860	1.80%
AR	1162	2694	34456	1.56%
RU	4691	1448	25830	1.56%
BR	2171	3476	61543	1.52%
Total	**36264**	**99677**	**907933**	**0.61%**

Game name	Refund to Completed payments (%)
Funny squirrel	0.56%
Funny squirrel in Space	0.90%
Total	**0.61%**

User country	Refund to Completed payments (%)
AD	1.89%
AE	0.39%
AF	0.00%
AG	0.00%
AI	0.00%
AL	0.79%
AM	0.28%
AN	0.00%
AO	0.00%
AR	1.56%
AT	0.30%
AU	0.47%
AW	0.35%
Total	**0.61%**

This section presents a focused analysis of refund behavior relative to successful transactions, highlighting patterns that may suggest dissatisfaction, payment friction, or operational weaknesses across games, countries, and payment systems.

☑ Global Refund Rate Overview

Across the entire dataset, the overall refund-to-completed payment ratio stands at 0.61%. This low percentage indicates a relatively healthy payment ecosystem, with minimal post-purchase reversal or dissatisfaction on a macro level.

☑ Game-Level Insights

Funny Squirrel in Space exhibits a higher refund rate (0.90%) compared to Funny Squirrel (0.56%).

While both are within acceptable thresholds, the discrepancy may suggest a slightly weaker post-payment experience or content satisfaction in the space-themed variant.

This gap should be further investigated, especially in months or countries where refund spikes are observed.

☑ Country-Level Observations

Zimbabwe (ZW) reports an alarming 17.39% refund rate, followed by Gabon (5.00%) and Saint Vincent and the Grenadines (3.85%).

Such high refund rates in low-volume countries often indicate either:

Localized technical or UX issues,

Fraudulent behavior or chargeback abuse,

Or cultural mismatch in transaction expectations.

Proactive investigation or regional payment optimization is recommended in these regions.

☑ Payment System Highlights

Most payment systems remain within expected limits, maintaining refund ratios under 20%.

However, Athleta shows a critical refund rate of 50%.

This is likely due to either extremely low transaction volume or misintegration.

It is strongly advised to audit this payment method's workflow and error handling mechanisms.

☑ Conclusion & Strategic Actions

Track refund behavior monthly to identify unusual activity early.

Segment refund causes by platform, system, or region for root cause analysis.

Reassess partnerships with underperforming payment providers to avoid recurring friction.

Enhance user support and refund communication flows in high-risk countries.

☑ Understanding refund ratios not only helps prevent revenue loss but also reflects how users perceive product quality and trust payment security.

10. Best Performing Payment Systems – UniPay e-wallet (new), WeChat

This KPI visual highlights the top-performing payment systems based on key reliability indicators such as:

- Lowest payment declines rates
- Minimal or zero refund ratios
- Consistent monthly stability
- No critical performance spikes

☑ Why UniPay e-wallet and WeChat?

1. UniPay e-wallet (new)

- Demonstrated zero or near-zero payment rejections.
- Maintained a stable refund rate across all observed months.
- Despite being newly introduced, it has shown exceptional transactional efficiency and can be scaled further.

2. WeChat

- Exhibited one of the lowest PS Decline rates in the dataset.
- Popular in multiple regions and capable of handling high traffic with minimal technical interruptions.
- Shows strong compatibility with the user base, especially in East Asian markets.

☑ Strategic Interpretation:

- These systems can serve as model payment providers for onboarding in future markets.
- Their performance reinforces user trust, contributes to higher conversion rates, and reduces support incidents related to failed payments.
- Promotion of these gateways in new or underperforming regions could help offset risks from high-failure systems.

☑ The identification of UniPay e-wallet (new) and WeChat as top systems reflects a data-driven approach to optimizing the payment infrastructure and elevating the user experience across global markets.

11. Payment System Performance Summary – Risk & Reliability Overview

This consolidated table evaluates each payment system's reliability using three main indicators:

1. Decline Rates (Payment System and AFS)
2. Refund Rate (%)
3. Status Flag: "Problematic" vs "OK"

☑ Key Performance Insights:

1. Systems flagged as Problematic:
    - A total of 24 out of 28 systems are marked as "Problematic", indicating issues such as:
    - High PS decline rates (>40%)

- Elevated refund rates
- Combination of multiple failure types
- Examples:
- Athleta: PS decline rate = 83.33%, refund rate = 50%
- Mobuy Pin&Call Germany: 78.53% decline rate, 2.71% refund
- Bank Transfers EU: 83.91% decline, 2.23% refund

2. Best-performing systems marked “OK”:
    - QIWI, Visa Checkout, Onebip Global PSMS, Bank SGB Poland, Xsolla Balance all passed thresholds with:
    - Decline rates under 40%
    - Refund rates under 5%
    - Low AFS interference
    - These systems are suitable for default routing in sensitive geographies.
3. High Decline, Low Refund (Risky Systems):
    - Some systems, like DnB Nord Bank or GameStop, have relatively low refund rates but high decline rates, which still affect user experience negatively.
    - These are still flagged as “Problematic” despite fewer chargebacks.

☑ Strategic Implications:

- Payment systems with combined high decline and refund rates create friction, reduce user trust, and lead to higher churn.
- Even low-usage systems like Athleta can disproportionately harm experience due to small transaction base and extreme failure ratios.
- Monitoring, testing, and reconsidering partnerships with such systems is crucial to optimize payment flow.

☑ Recommendations:

- Immediately review high-risk systems (Decline Rate > 50% or Refund Rate > 10%)
- Consider replacing or deprioritizing:
- Athleta, Mobuy Pin&Call, Bank Transfers EU, PagSeguro
- Promote “OK” systems like QIWI and Visa Checkout in primary user journeys
- Apply stricter fraud detection tuning and fallback options in high-failure platforms

☑ This performance breakdown acts as a reliability dashboard, enabling the business and technical teams to make data-driven decisions about payment routing, system prioritization, and end-user experience.